Principal Component Analysis May Not Always be Useful When Working in High Dimensions*

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1 Introduction

In statistics, multivariate analysis may seem redundant. In many cases, it makes just as much sense to perform a univariate analysis alone or with some correction. For example, univariate analysis of variance (ANOVA) may give us more valuable information than multivariate ANOVA. As such, it is fitting to ask in which cases multivariate analysis is more useful than univariate analysis. In the age of "big data", being able to reduce the dimensionality of a dataset has been popular through methods such as principal component analysis (PCA) and singular value decomposition (SVD). This study, in particular, explores univariate linear regression and PCA. Specifically, I aim to answer the following research question: How are predictions affected when principal components are used as covariates instead of the data themselves?

To answer this research question, I employ data from the Agency for Healthcare Research and Quality (AHRQ) database to make predictions about income using households, people, internet connectivity, educational attainment, housing and mobility, transportation, health insurance status, and environment variables as predictors. I then create a univariate multiple regression model with all of the predictors and calculate the mean squared prediction error (MSPE) on the validation and test set. A second model is created using the top 26 principal components that explained the most variance in the data matrix as predictors. The MSPE is also calculated for this model and compared to the first model.

Through this, I found that the univariate model performs much better than the second model. While both models have a high MSPE, the univariate model's is more than half that of the second model. This shows that dimension reduction may not always be useful and in fact make results less interpretable.

*Project repository available at: https://github.com/meganajoseph/pca_reg_comp.

The rest of the paper is structured as follows: Section 2 discusses the data, Section 3 discusses the models, Section 4 presents the results, and Section 5 discusses the implications of the results.

2 Data

The dataset comes from the Agency for Healthcare Research and Quality (AHRQ) database (Agency for Healthcare Research and Quality 2025a). They compile data from multiple sources, but the sources of interest are the American Community Survey (ACS) and Community Resilience Estimates (CRE) (Agency for Healthcare Research and Quality 2025b). The U.S. Census Bureau conducts the ACS in order to collect social, economic, housing, and demographic data across the United States (Agency for Healthcare Research and Quality 2025b). The CRE aims to capture community resilience, and variables are constructed from the ACS and Population Estimates Program (PEP) (Agency for Healthcare Research and Quality 2025b).

Each row represents a census tract in the United States in 2023, and there are 85433 rows. The 46 variables of interest can be split into categories: households, people, internet connectivity, educational attainment, housing and mobility, transportation, health insurance status, environment, and income. The households variables are average household size and percentage of limited English speaking households. Some of the people variables are percentage of population with a disability, percentage of population that is male, percentage of population who are not U.S. citizens, percentage of population ages 5 and over that only speaks English, percentage of population ages 5 and over that speaks other languages, percentage of civilian population ages 18 and over consisting of veterans, median age of total population, percentage of population ages 65 and over, and the percentage of families with children that are single-parent families. The remaining people variables describe race, ethnicity, and marital status. Race is measured as the percentage of the population reporting each of the following races alone and in combination with one or more other races: American Indian and Alaska Native, Asian, Black or African American, Native Hawaiian and Pacific Islander, White, and some other race. Ethnicity and multiracial identity are captured by the percentage of the population reporting Hispanic ethnicity and the percentage reporting multiple races. Marital status is recorded separately for the female and male populations ages 15 and over as the percentages that are divorced or separated, now married with spouse absent, now married with spouse present, never married, and now widowed. The internet connectivity variables are the percentage of households without a computing device, percentage of households with a smartphone, percentage of households with any internet subscription, and percentage of households with a cellular data plan. Educational attainment is measured by the percentage of the population ages 25 and over with a bachelor's degree. The housing and mobility variables are the percentage of housing units with more than one occupant per room, percentage of occupied housing units with solar energy, and median year structure built of housing units. Transportation is measured by the percentage of housing units with no vehicle available. The health insurance status variable is the percentage of the population with no health insurance coverage. The environment variable is the estimated number of individuals

with zero risk factors (from the Community Resilience Estimates). Finally, income is measured by median household income, in dollars adjusted for inflation to the data file year, and this is the response variable. Predictors that are correlated with income, such as enrollment in government assistance programs, were not included.

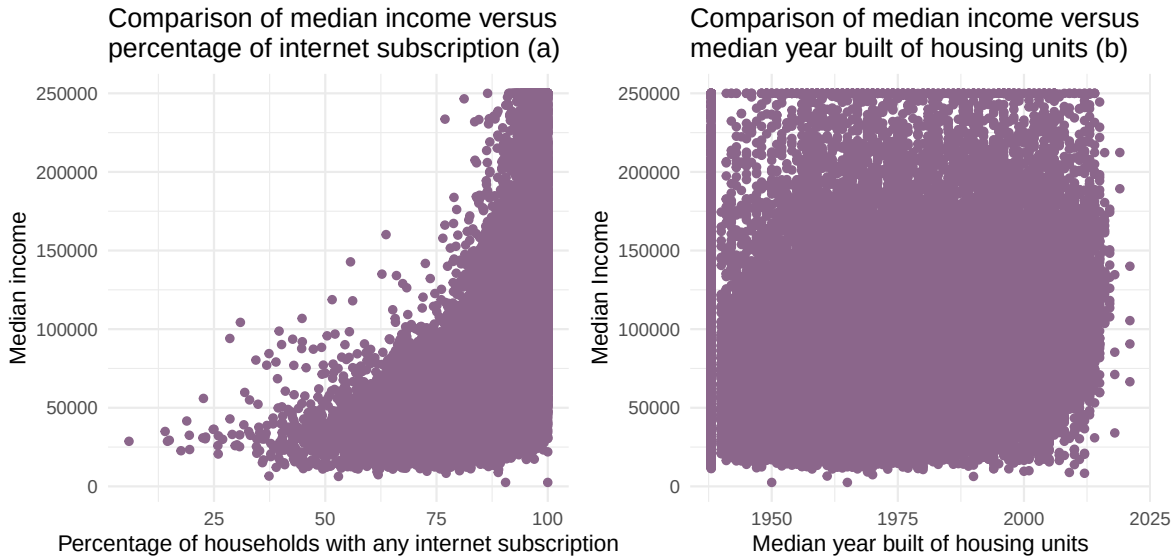


Figure 1: Plot (a) shows a scatter plot showing the positive, non-linear relationship between the percentage of households with any internet subscription (x-axis) and median household income (y-axis). Plot (b) shows a scatter plot showing little relationship between the median year housing units were built in (x-axis) and median income (y-axis).

Most of the variables have what looks more like a non-linear relationship with income. For example, Figure 1 (a) shows a positive relationship between the percentage of households in a census tract with any internet subscription and the median household income. However, it looks more exponential than linear. There are also a few variables with little relationship with income. The median year housing units were built in a census tract is one such example. Figure 1 (b) shows this where the points cover the entire area in no recognizable shape.

There are 47815 NA values in the dataset. After dropping those values, the dataset include 82272 rows.

3 Methods

3.1 Univariate

The univariate method will be multiple linear regression with median income as the response and the rest of the variables are predictors. For simplicity, interactions will not be included, and transformations will not be applied even though they may improve performance. Let Y be the 82272 x 1 vector where each entry represents the median income in a census tract. Let X be the 82272 x 46 matrix where the first column is the vector of ones and the rest represent one predictor. The model can be written as

$$Y = X\beta + \varepsilon$$

where β is the vector of coefficients for each predictor, and ε is the vector containing the error. Here, each β represents the average change in median income in a census tract in the United States for a unit increase in the corresponding predictor while holding the other predictors constant.

3.1.1 Assumptions of Linear Regression

The assumptions of using linear regression for inference are:

1. The variables used accurately reflect the quantities of interest, the model should include all predictors, and the model should be applicable for the research question.
2. The sample data is representative of the population of interest.
3. The mean function must be correct.
4. Errors must be independent.
5. The variance of the errors must be equal.
6. Errors must be normal.

Since the goal of this analysis is prediction, it is not necessary for the assumptions to be met.

3.2 Multivariate

PCA is the multivariate method of choice for this study. PCA is applied in order to project p-dimensional data onto a k-dimensional space (where $k <$ p) that has maximum variance. It works by choosing the unit vector v such that the mean squared error (MSE) is minimized when the data X is projected onto v. This is the same as finding the v that maximizes the sample variance of the projections onto v:

$$\max_{v \in \mathbb{R}^p, ||v||=1} v^T \hat{S} v$$

whose solution is the eigenvector corresponding to the eigenvalue λ_1 of $\hat{S}$ where $\hat{S}$ is the sample covariance of X.

The output of PCA is the set of maximum-variance principal directions which can be used to find the principal components by projecting the data matrix X onto the first k principal directions. The components can be interpreted as the new coordinates of each observation in the new basis.

Once PCA is applied to the centered predictors, the top k components that capture approximately 95% of the variance will be chosen. The k loadings will be multiplied with the predictor matrix and used as the new predictors for the same regression method described previously.

Suppose V_k is a matrix containing the k variable loadings, and $X^* = XV_K$ is the matrix containing the new coordinates after projecting the data onto the k-dimensional space. Then, we can define the model:

$$Y = X^*\beta + \varepsilon$$

A scree plot can help determine the number of components to use based on some requirement. I choose to pick the k components that explain at least 95% of the variance. This plot shows how much variance is explained by each component.

The main difference between these two methods is what is being used as predictors in the regression model. The univariate case simply uses all of the variables as is. The PCA method uses the principal components as the predictors. Additionally, there should be less predictors in the PCA method than the univariate method since the aim is the reduce the dimensionality.

3.3 Comparing the Models

To compare the models, I will first split the data into 70% training, 15% testing, and 15% validation sets. Then, I will train the models on the training set and find the root mean squared prediction error using the validation set. This is defined as

$$\sqrt{E[(Y^* - \hat{Y}^*)^2]}$$

where Y^* is the true median income and $\hat{Y}^*$ is the predicted median income. This is the square root of the expected squared difference between the prediction and the true value over the validation data population. The one that gives the lowest root mean squared error will be the best model. Finally, I will use the testing set to estimate the performance on the better model.

I would expect that the PCA method would perform better because it selects the directions that capture the highest variance, and linear regression may be more effective when the predictors capture more of the variance in the response. However, it would not be unexpected if they performed about the same since PCA is reducing the number of dimensions which may remove important information when explaining the response.

3.4 Software

I use the `lm()` function from the R programming language (R Core Team 2025) to create the univariate model and the PCA model. I also use the `prcomp()` function to perform PCA on the data matrix.

4 Results

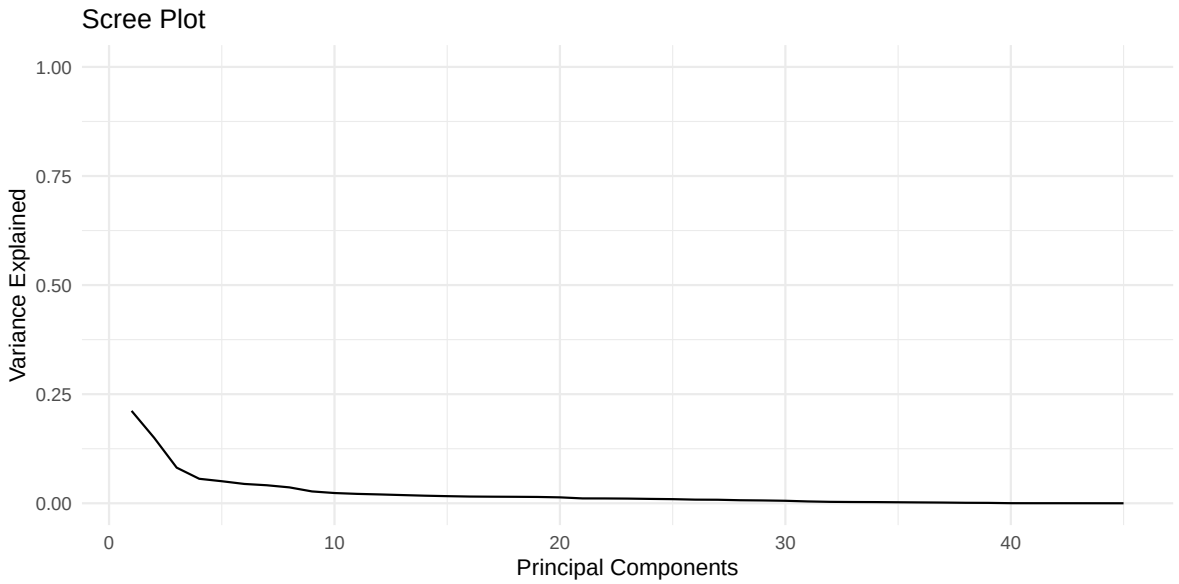


Figure 2: This plot shows the variance explained by each component after PCA is performed on the predictor variables.

The scree plot developed is shown in Figure 2. Based on this, I chose to pick the top 26 components which reduced the model to a little over half the number of predictors compared to the univariate model. The components don't seem to explain much of the variance, with the first component not even reaching 25% of the variance.

Appendix A shows the slope estimate values for the univariate method and the multivariate method. The β values can be interpreted as the average change in median income for a unit increase in the principal components, but they are not as interpretable as the univariate coefficients. The values vary between the two, even with opposite signs.

The model with the lowest RMSPE is the univariate model with all of the predictors. Table 1 shows the RMSPE values for each model based on predictions from the validation set and for the better model on the testing set.

Table 1: This table shows the root mean squared prediction errors on the validation set for each model and for the best model on the testing set.

Univariate Model RMSPE (Validation)	PCA Model RMSPE (Validation)	Final Model RMSPE (Test)
20772.91	53102.352	20662.813

However, both errors are quite large which can be attributed to many reasons such as the non-linearity mentioned earlier, lack of interaction terms, and/or more intricate nuances surrounding income in a census tract.

These results show that reducing the dimensionality of a data matrix may not always be helpful in prediction. In this case, the univariate error was more than half that of the multivariate error. The principal components themselves did not explain much of the variance in the data which may also contribute to the poor performance. Applying PCA also reduced the ineterpretability of the model which may be important in certain use cases. This shows that when creating models, it's important base decisions off of theory and what we know about how the predictors are related to the response.

5 Discussion

The aim of this study was to answer the following research question: How does using principal components as predictors affect predictions from a multiple regression model? I investigated this using data from the Agency for Healthcare Research and Quality. Then, I created a univariate multiple linear regression model using all of the predictors without interaction terms and another multiple regression model using the top 26 principal components as predictors. The model that predicted the best was the univariate model with a test root mean squared prediction error of 20662.813.

5.1 Potential Weaknesses

One weakness of this study is the lack of interaction terms. Including them may impact how much variance the principal components explained and lead to better results. They may also enhance the univariate model as well.

Another weakness is that transformations should have been applied to the variables. Many of the variables appeared to have a non-linear association with median income. Applying an exponential, log, or another relevant transformation may have improved the prediction error in both models.

5.2 Possible Extensions

An extension could be to address the weaknesses proposed earlier to see how they affect prediction performance. Another extension could be to perform the study at the individual level rather than at the census tract level to see if demographics can predict a person's income.

6 References

Agency for Healthcare Research and Quality. 2025a. *AHRQ Community-Level Health (CLH) Database: 2023 Census Tract Data*. Https://www.ahrq.gov/data/innovations/clh-data.html; Agency for Healthcare Research; Quality.

Agency for Healthcare Research and Quality. 2025b. *AHRQ Community-Level Health (CLH) Database: Data Source Documentation*. Data Source Documentation. Agency for Healthcare Research; Quality. https://www.ahrq.gov/sites/default/files/wysiwyg/data/clh/CLH-Data-Sources-Documentation-2025.pdf.

R Core Team. 2025. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. https://www.R-project.org/.

A Slope Estimates

Slope Estimate	Univariate Value	PCA Value
$\hat{\beta}_0$	875327.27	230180.685
$\hat{\beta}_1$	23320.025	508.311
$\hat{\beta}_2$	-829.694	1168.995
$\hat{\beta}_3$	289.505	-309.189
$\hat{\beta}_4$	197.918	434.843
$\hat{\beta}_5$	-54.739	146.446
$\hat{\beta}_6$	-699.738	206.751
$\hat{\beta}_7$	-169.843	111.208
$\hat{\beta}_8$	-430.597	-23.791
$\hat{\beta}_9$	1416.178	20.86
$\hat{\beta}_{10}$	-397.486	-725.682
$\hat{\beta}_{11}$	-7677.912	1244.799
$\hat{\beta}_{12}$	-9221.631	-787.622
$\hat{\beta}_{13}$	-7858.161	-1266.86

Slope Estimate	Univariate Value	PCA Value
$\hat{\beta}_{14}$	-8.922	-683.345
$\hat{\beta}_{15}$	-8288.343	-15.541
$\hat{\beta}_{16}$	-6576.032	-300.496
$\hat{\beta}_{17}$	-8072.924	801.527
$\hat{\beta}_{18}$	-8394.725	-139.116
$\hat{\beta}_{19}$	-55.973	406.625
$\hat{\beta}_{20}$	-120.84	-948.021
$\hat{\beta}_{21}$	1514.855	46.352
$\hat{\beta}_{22}$	-176.576	-24.623
$\hat{\beta}_{23}$	-1487.778	-367.522
$\hat{\beta}_{24}$	365.607	140.019
$\hat{\beta}_{25}$	325.388	391.508
$\hat{\beta}_{26}$	449.107	-194.551
$\hat{\beta}_{27}$	90.665	NA
$\hat{\beta}_{28}$	374.312	NA
$\hat{\beta}_{29}$	1262.032	NA
$\hat{\beta}_{30}$	-731.661	NA
$\hat{\beta}_{31}$	740.215	NA
$\hat{\beta}_{32}$	-127.395	NA
$\hat{\beta}_{33}$	185.137	NA
$\hat{\beta}_{34}$	-490.885	NA
$\hat{\beta}_{35}$	3.199	NA
$\hat{\beta}_{36}$	-28.454	NA
$\hat{\beta}_{37}$	43.525	NA
$\hat{\beta}_{38}$	343.566	NA
$\hat{\beta}_{39}$	119.538	NA
$\hat{\beta}_{40}$	390.337	NA
$\hat{\beta}_{41}$	414.781	NA
$\hat{\beta}_{42}$	223.105	NA
$\hat{\beta}_{43}$	167.448	NA
$\hat{\beta}_{44}$	56.907	NA
$\hat{\beta}_{45}$	161.355	NA